

Manchester encoding is a **line coding technique** used in digital communication to represent binary data (0s and 1s) in a way that ensures **synchronization between sender and receiver**. It is widely used in Ethernet (IEEE 802.3), RFID, and other communication systems.

◇ Definition:

Manchester encoding represents each bit of data with a **transition in the middle of the bit period**, ensuring that the clock and data are synchronized.

- **Logic '1'** → Transition from **low to high** ($\downarrow\uparrow$).
 - **Logic '0'** → Transition from **high to low** ($\uparrow\downarrow$).
- (Note: Some standards use the opposite convention, but the mid-bit transition rule is always there.)*
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◇ Characteristics:

1. **Self-clocking** → The mid-bit transition provides a clock signal, so no separate clock line is needed.
 2. **No DC component** → Average voltage stays around zero.
 3. **Bandwidth requirement** → Requires **double the bandwidth** compared to NRZ (Non-Return to Zero).
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◇ Example:

For the bit stream: 1 0 1 1 0

- 1 → Low → High
- 0 → High → Low
- 1 → Low → High
- 1 → Low → High
- 0 → High → Low

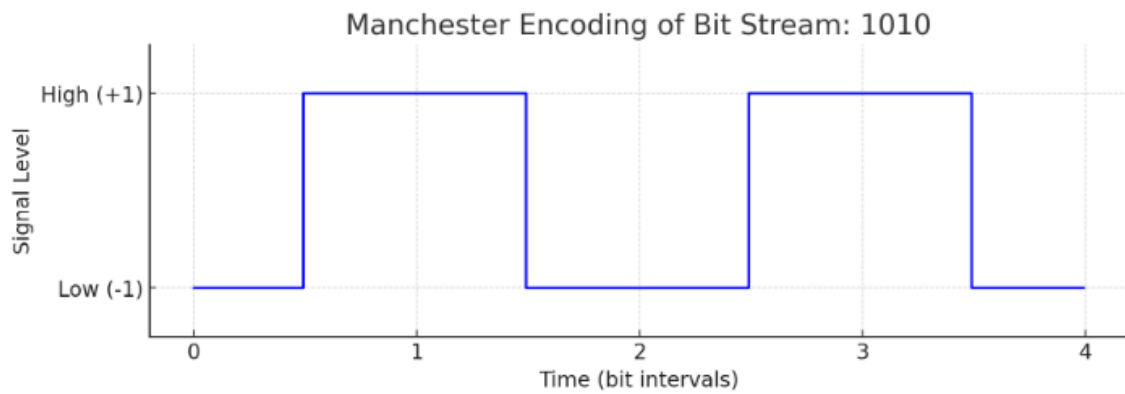
Waveform pattern would look like alternating transitions at the middle of each bit.

◇ Advantages:

- Easy clock recovery.
- Reduces synchronization errors.
- Eliminates long strings of 0s or 1s (no loss of transitions).

◇ Disadvantages:

- Consumes more bandwidth than NRZ.
- More complex decoding compared to simpler schemes.



Here's the **Manchester encoded waveform** for the bit stream 1010 :

- Each bit has a **transition in the middle**.
- 1 → Low-to-High transition.
- 0 → High-to-Low transition.

